Lab Work 2: AIOps Detection Engine - Engineering Report

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Objective: The objective of this lab is to extend the monitoring pipeline by implementing an AIOps Detection Engine in Laravel. The engine aims to proactively pull telemetry data, learn system baselines, detect anomalies using multi-signal correlation, and trigger structured alerts.

1. **Detection Command & Metrics Integration** The engine was implemented as a robust Artisan command (aiops:detect). It runs continuously using the --daemon flag, polling data from the Prometheus API every 10 seconds. It successfully parses PromQL queries for Traffic, Latency, and Error rates.

```
PS C:\aiops-lab> php artisan aiops:detect --daemon
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Normal - T: 0.02, L: 5, E: 0.017
System Normal - T: 0.22, L: 5, E: 0.043
```

2. **Baseline Modeling (Database Persistence)** To establish a reference for "normal" system behavior, the engine calculates baselines during a predefined learning window. To ensure data persistence and statefulness, this baseline is saved in a SQLite database table (aiops_baselines) capturing the metric name, calculated value, and the explicit window_start and window_end timestamps.

database > database.sqlite

Rows: 1

	id	metric_n...	value	window_start	window_end	created_at	updated_at
1	1	traffic	1.6957406473381	2026-03-11 21:...	2026-03-11 21:...	2026-03-11 21:...	2026-03-11 21:...
2							

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3. **Anomaly Detection, Event Correlation & Alerting** The system continuously compares incoming metrics against the stored baselines. Upon detecting deviations (e.g., an error rate spike exceeding the threshold), the correlation logic maps the combined signals to a specific Root Cause. As demonstrated below, the system successfully identified an APPLICATION_BUG and outputted a properly formatted alert containing the Timestamp, Severity Level, Summary, and a uniquely generated Incident ID. A Laravel Event (IncidentCreated) was also triggered programmatically.

```
PS C:\aiops-lab> php artisan aiops:detect --daemon
System Idle - Waiting for data...
System Idle - Waiting for data...
System Idle - Waiting for data...
System Normal - T: 0.02, L: 5, E: 0.017
System Normal - T: 0.22, L: 5, E: 0.043
System Normal - T: 0.3, L: 5, E: 0.043
[22:12:56] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-GK2L6J)
[22:13:06] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-XBJ9RO)
[22:13:16] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-VKHW02)
[22:13:26] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-WOK0TS)
[22:13:36] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-S1LQMI)
[22:13:46] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-ECI1TH)
```

```
[22:14:47] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-KBSTIP)
[22:14:57] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-KG021W)
[22:15:07] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-YRGKJB)
[22:15:17] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-WMSTUB)
[22:15:27] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-KP3PMT)
[22:15:37] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-PZDONZ)
[22:15:47] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-ASNOAE)
[22:15:57] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-CC6GCL)
[22:16:07] [ALERT] [WARNING] Anomaly Detected: APPLICATION_BUG (ID: INC-DJZEEF)
```

```
php
python
powershell
php
```

4. **Structured Incident Generation** For every detected anomaly, the engine constructs a structured JSON object representing the incident. This ensures that incident data (ID, Timestamp, Root Cause, Severity, and Metric Signals) is machine-readable and ready for integration with external IT Service Management (ITSM) tools.


```
},
{
    "incident_id": "INC-DLX41R",
    "timestamp": "2026-03-11T20:45:00+00:00",
    "root_cause": "APPLICATION_BUG \/ ANOMALY_DETECTED",
    "severity": "CRITICAL",
    "signals": {
        "traffic": 0.5,
        "latency": 5,
        "errors": 0.052
    },
    "status": "OPEN"
},
{
    "incident_id": "INC-J3TFP3",
    "timestamp": "2026-03-11T20:45:05+00:00",
    "root_cause": "APPLICATION_BUG \/ ANOMALY_DETECTED",
    "severity": "CRITICAL",
    "signals": {
        "traffic": 0.44,
        "latency": 5,
        "errors": 0.052
    },
    "status": "OPEN"
},
{
    "incident_id": "INC-K9ALAV",
    "timestamp": "2026-03-11T21:08:46+00:00",
    "root_cause": "APPLICATION_BUG",
    "severity": "WARNING",
    "signals": {
        "traffic": 1.7,
        "latency": 5,
        "errors": 0.183
    },
    "status": "OPEN"
},
{
    "incident_id": "INC-RV4ZSV",
```


Conclusion: All 7 laboratory requirements were successfully implemented. The Detection Engine effectively differentiates between idle states, normal loads, and anomalous behavior, establishing a reliable foundation for an AIOps pipeline.